LONG VU VAN

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EDUCATION

Vietnam National University (UET - VNU) **Cau Giay, Ha Noi, Vietnam**
Major: Artificial Intelligence *2022 - 2026*

- Graduated with **High Distinction** in Artificial Intelligence, with a cumulative GPA of 3.74/4.00, ranked top 5 of the graduating cohort.
- Selected as one of 28 students from a cohort of 174 for the Advanced Program in Artificial Intelligence.
- Ranked 1st out of 174 students in Semester I of academic year 2023-2024.

High School for Gifted Students (Chuyên Sư phạm) **Cau Giay, Ha Noi, Vietnam**
Third Prize, Vietnam National Olympiad in Informatics *2021*

TECHNICAL SKILLS

Programming Languages: Python, C++, Java
Frameworks & Libraries: PyTorch, TensorFlow, Hugging Face, Scikit-learn, NumPy, Matplotlib, Seaborn, PySpark
Tools: Git, GitHub, Docker
Deep Learning: Specialized in Diffusion Models, Video Generation Models and World Models.
Robotics: Confident with robot-learning frameworks, especially Vision-Language-Action models (VLAs) : openpi (π_0, $\pi_{0.5}$), NVIDIA GR00T (GR00T 1.5, GR00T 1.7) and world models.
Languages: Vietnamese (native), English (professional working proficiency)

WORK EXPERIENCE

AI Resident **6/2025 - Present**
Qualcomm AI Residency Program, Hanoi, Vietnam

- Conduct research on top-tier video generation models, world models, and robotic manipulation within a competitive, selective residency program.
- Develop and adapt frontier generative and world-modeling techniques to advance data-driven approaches for robot learning.

AI Engineer **6/2024 - 11/2024**
Oraichain Lab, Hanoi, Vietnam

- Researched and developed human-animation video generation models.
- Fine-tuned and optimized models for compatibility with Oraichain's infrastructure.

Research Student **12/2022 - 5/2023**
IAI-UET, Hanoi, Vietnam

- Tested and developed state-of-the-art models and validated research hypotheses.
- Containerized models with Docker to streamline reproducibility and future deployment.

PROJECTS

UniCaMo: Unified Camera and Motion Control for Video Diffusion Models
AI Research

- Proposed a 3D-grounded motion-consistent noise representation that jointly controls object trajectories and camera viewpoints in image-to-video diffusion purely by constructing the initial noise — no adapters or architectural changes, only lightweight LoRA fine-tuning of Wan2.1-14B models.
- Designed the two core components: **Track-the-Noise** (warps first-frame noise along 3D point tracks with depth-aware collision handling) and **Move-the-World** (view-consistent sphere-projected noise for regions revealed by camera motion).
- Achieved state-of-the-art quality and controllability on MoveBench (e.g., FID 10.36 / FVD 78.6 / EPE 2.30 vs. Wan-Move's 12.23 / 83.5 / 2.60); first-author submission under review at ICLR 2027.

RoboDream: High-Quality Data Generation for Robotic Manipulation
AI Research

- Built an end-to-end pipeline that synthesizes new manipulation episodes for small robotics datasets using an action-conditioned world model: world-model fine-tuning, policy-in-the-loop imagined rollouts, and data filtering, targeting VLA policies such as $\pi_{0.5}$.

- Demonstrated in MolmoSpaces simulation that fine-tuning $\pi_{0.5}$ on the synthesized data improves pick-and-place success rate from 17.9% (fine-tuned on the original small real dataset; base $\pi_{0.5}$-DROID: 13.6%) to **23.1%** (fine-tuned on the real data plus synthesized data). **Improve $\pi_{0.5}$ task success rate from 45% to 55% in real robot execution.**

Image Quality Assessment-Driven Learning **Github**

AI Research

- Applied **StarGAN** for image generation combined with a novel cross-domain learning method to achieve state-of-the-art performance.
- Designed a new loss function for StarGAN to mitigate the impact of low-quality generated images during training.
- Improved **mAP** by **2.3** points and **Rank-1** accuracy by **3.4** points.

Clustering Images by Color for Big Data **Github**

AI Engineer

- Leveraged **PySpark** with the **K-means** algorithm to process and cluster large-scale image data.
- Configured and managed **Tailscale** to establish secure connections between nodes, enabling efficient distributed task allocation.
- Visualized and evaluated clustering results using **Matplotlib**.

ComposeAE - Image Retrieval **Github**

AI Engineer

- Implemented a Dockerfile to improve the reproducibility of experiments.
- Developed single-sample inference code for an e-commerce retrieval model.
- Retrained the model to achieve improved results on the Fashion200k dataset (23.1 Rank-1 score).

ACTIVITIES & AWARDS

- **Encouragement Scholarship**, Semester I (2023-2024, 2024-2025).
- **Second Prize**, ACM/ICPC Vietnam National Contest, 2022, 2023, and 2024.
- **Third Prize**, Vietnam National Olympiad in Informatics, 2021.
- **Bronze Medal**, ICPC Asia Regional Contest, 2024.
- **First Prize**, Scientific Research Conference for Undergraduates, 2024.

PUBLICATIONS

- IQAGA: Image Quality Assessment-Driven Learning with GAN-Based Dataset Augmentation for Cross-Domain Person Re-Identification — SOICT 2023 Link
- UniCaMo: Unified Camera and Motion Control for Video Diffusion Models. Submitted to ICLR 2027 (under review). ArXiv Preprint